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ECE 432 Digital VLSI Bonus Project

Traffic Light Controller

under supervision of Prof. Mohammed Sharaf

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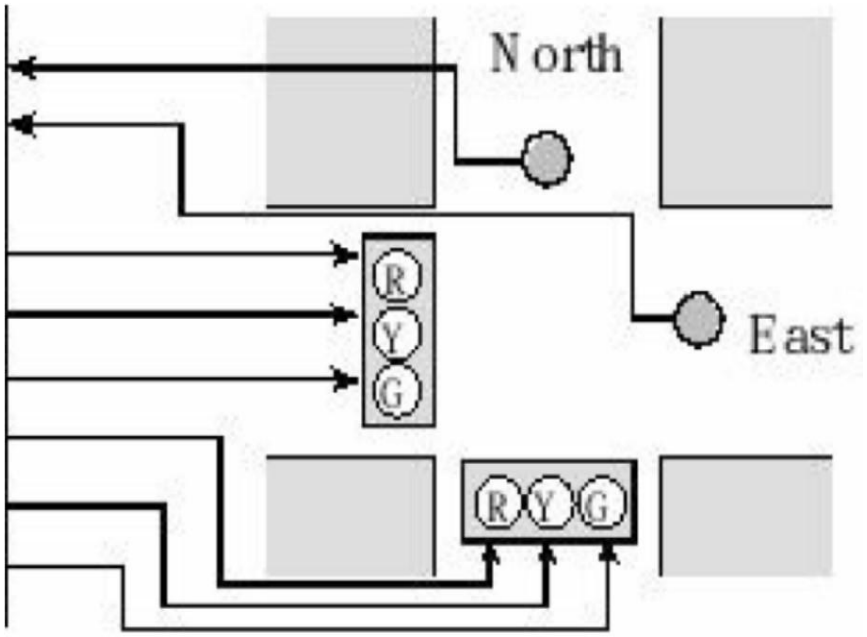
Tools: Quartus, ModelSim, Cyclone III EP3C120F780C7 FPGA

Project Overview

This project implements a traffic light controller for an intersection of two one-way streets (North and East). The controller uses car presence sensors (simulated via FPGA switches) to dynamically manage traffic flow. The design is fully compliant with the specified behavioral requirements and is implemented on an Intel Cyclone III EP3C120F780C7 FPGA.



The controller must satisfy the following conditions:

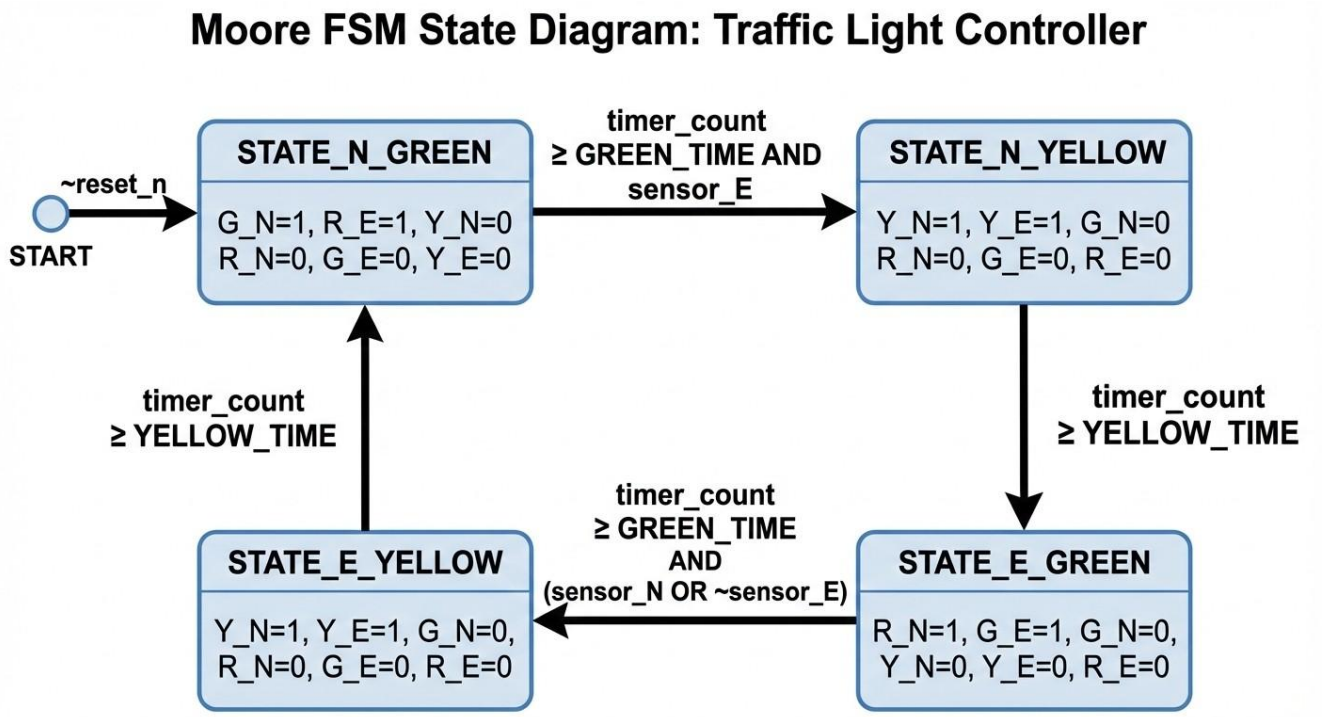
- If **no cars are present**, remain in **any green state indefinitely**.
- **Green phases** last **at least 30 seconds**.
- **Yellow transition** lasts **exactly 5 seconds** with **both directions yellow**.
- If cars are present on **only one road**, stay green on that road

If cars are present on **both roads**, cycle through:

1. Green North – Red East (30 s)
2. Yellow North – Yellow East (5 s)
3. Red North – Green East (30 s)
4. Yellow North – Yellow East (5 s)

Additionally, the system must:

- Use **two switches** as car sensors.
- Drive **six LEDs** (GN, YN, RN, GE, YE, RE).
- Display the **real-time countdown (in seconds)** on a **dual-digit 7-segment display**.



The system is built using a **synchronous finite state machine (FSM)** with four states:

- `STATE_N_GREEN`: North green, East red
- `STATE_N_YELLOW`: Both yellow (N → E transition)
- `STATE_E_GREEN`: East green, North red
- `STATE_E_YELLOW`: Both yellow (E → N transition)

Verification Plan

The design was verified through simulation test scenarios in ModelSim and real-time operation on the FPGA, with the test cases in the table below covering all functional requirements at the simulation level.

Test Scenario	Initial State	Expected Behavior	Pass Criteria
No cars present	North Green	Remain in North Green indefinitely	`G_N = 1`, `R_E = 1` after >30s
Only North car	North Green	Stay in North Green	`G_N = 1`, `R_E = 1` after >30s
Only East car	North Green	Transition to East Green after 30s + 5s yellow	`R_N = 1`, `G_E = 1`
Both cars (full cycle)	Reset (North Green)	Complete full 70s cycle: N-green → both-yellow → E-green → both-yellow → N-green	Observe all 4 states in sequence
East car leaves during North green	North Green	Remain in North Green after East car departs	`G_N = 1`, `R_E = 1`
Critical: No cars in East Green	East Green	**Remain in East Green**	`G_E = 1`, `R_N = 1` after >30s

Code Implementation

Design: traffic_light_controller

```
module traffic_light_controller
#(parameter GREEN_TIME = 30, YELLOW_TIME = 5, CLK_DIV = 50000000, MUX_CLK_DIV = 50000)
(
  input clk,
  input reset_n,
  input sensor_N,
  input sensor_E,
  output G_N,
  output Y_N,
  output R_N,
  output G_E,
  output Y_E,
  output R_E,
  output [6:0] seg,
  output [1:0] digit_sel
);

  localparam STATE_N_GREEN   = 2'b00,
             STATE_N_YELLOW  = 2'b01,
             STATE_E_GREEN   = 2'b10,
             STATE_E_YELLOW  = 2'b11;

  reg [1:0] state_reg, state_next;
  reg [1:0] prev_state;

  wire clk_1hz;
  wire  clk_mux;
  reg digit_toggle;

  wire timer_done;
  reg timer_start;
  wire [5:0] timer_count;
  reg [5:0] timer_target;

  wire [5:0] display_time;
  wire [3:0] ones_digit, tens_digit;
  wire [3:0] current_digit;
  wire [6:0] seg_data;

  clock_divider #(.DIV_VALUE(CLK_DIV)) clk_div(
    .clk_in(clk),
    .reset_n(reset_n),
    .clk_out(clk_1hz)
  );

  mux_clock_divider #(.DIV_VALUE(MUX_CLK_DIV)) mux_clk_div(
    .clk_in(clk),
    .reset_n(reset_n),
    .clk_out(clk_mux)
  );

  timer #(.FINAL_VALUE(GREEN_TIME)) main_timer(
    .clk(clk_1hz),
    .reset_n(reset_n),
    .start(timer_start),
    .done(timer_done),
    .count(timer_count)
  );
```

```
always @(posedge clk_1hz, negedge reset_n)
begin
  if (~reset_n) begin
    state_reg <= STATE_N_GREEN;
    prev_state <= STATE_N_GREEN;
    timer_start <= 1'b1;
  end
  else begin
    prev_state <= state_reg;
    state_reg <= state_next;
    timer_start <= (state_next != state_reg);
  end
end

always @(*) begin
  case(state_reg)
    STATE_N_GREEN: timer_target = GREEN_TIME;
    STATE_N_YELLOW: timer_target = YELLOW_TIME;
    STATE_E_GREEN: timer_target = GREEN_TIME;
    STATE_E_YELLOW: timer_target = YELLOW_TIME;
    default:        timer_target = GREEN_TIME;
  endcase
end

always @(*)
begin
  state_next = state_reg;

  case(state_reg)
    STATE_N_GREEN: begin
      if (timer_count >= GREEN_TIME) begin
        if (sensor_E)
          state_next = STATE_N_YELLOW;
      end
    end

    STATE_N_YELLOW: begin
      if (timer_count >= YELLOW_TIME)
        state_next = STATE_E_GREEN;
    end

    STATE_E_GREEN: begin
      if (timer_count >= GREEN_TIME) begin
        if (sensor_N)
          state_next = STATE_E_YELLOW;
        else if (~sensor_E)
          state_next = STATE_E_YELLOW;
      end
    end

    STATE_E_YELLOW: begin
      if (timer_count >= YELLOW_TIME)
        state_next = STATE_N_GREEN;
    end

    default: state_next = STATE_N_GREEN;
  endcase
end

assign G_N = (state_reg == STATE_N_GREEN);
assign Y_N = (state_reg == STATE_N_YELLOW) || (state_reg == STATE_E_YELLOW);
assign R_N = (state_reg == STATE_E_GREEN);
```

```
  assign G_E = (state_reg == STATE_E_GREEN);
  assign Y_E = (state_reg == STATE_N_YELLOW) || (state_reg == STATE_E_YELLOW);
  assign R_E = (state_reg == STATE_N_GREEN);

  assign display_time = (state_reg == STATE_N_GREEN || state_reg == STATE_E_GREEN) ?
                        ((GREEN_TIME > timer_count) ? (GREEN_TIME - timer_count) : 6'd0) :
                        ((YELLOW_TIME > timer_count) ? (YELLOW_TIME - timer_count) : 6'd0);

  assign ones_digit = display_time % 10;
  assign tens_digit = display_time / 10;

  always @(posedge clk_mux, negedge reset_n)
  begin
    if (~reset_n)
      digit_toggle <= 1'b0;
    else
      digit_toggle <= ~digit_toggle;
  end

  assign current_digit = digit_toggle ? tens_digit : ones_digit;
  assign digit_sel = digit_toggle ? 2'b01 : 2'b10;

  seven_seg_decoder seg_decoder(
    .digit(current_digit),
    .seg(seg)
  );

endmodule
```

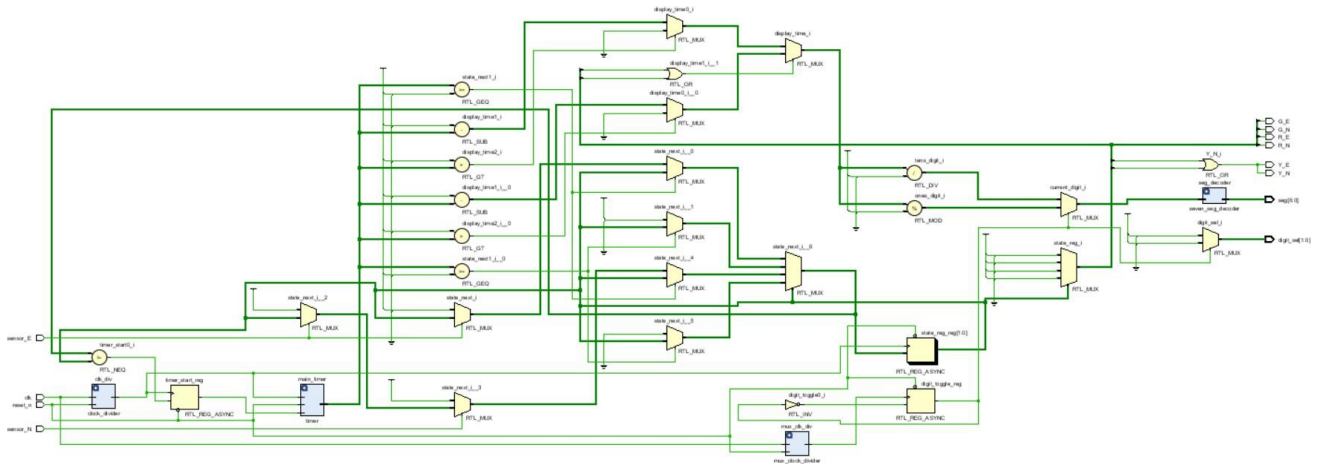
Design: timer

```
module timer
#(parameter FINAL_VALUE = 30)
(
  input clk,
  input reset_n,
  input start,
  output done,
  output [5:0] count
);

localparam BITS = 6;

reg [BITS-1:0] Q;
reg running;

always @(posedge clk or negedge reset_n)
begin
  if (~reset_n) begin
    Q <= 0;
    running <= 0;
  end
  else if (start) begin
    Q <= 0;
    running <= 1;
  end
  else if (running && ~done) begin
    Q <= Q + 1;
  end
  else if (done) begin
    running <= 0;
  end
end

assign done = (Q >= FINAL_VALUE);
assign count = Q;

endmodule

module clock_divider
#(parameter DIV_VALUE = 50000000)
(
  input clk_in,
  input reset_n,
  output reg clk_out
);

generate
  if (DIV_VALUE <= 1) begin : passthrough
    always @(posedge clk_in, negedge reset_n)
    begin
      if (~reset_n)
        clk_out <= 0;
      else
        clk_out <= ~clk_out;
    end
  end
  else begin : divider
    localparam BITS = $clog2(DIV_VALUE) + 1;
    reg [BITS-1:0] counter;
```

```
    always @(posedge clk_in, negedge reset_n)
    begin
      if (~reset_n) begin
        counter <= 0;
        clk_out <= 0;
      end
      else if (counter >= (DIV_VALUE/2 - 1)) begin
        counter <= 0;
        clk_out <= ~clk_out;
      end
      else begin
        counter <= counter + 1;
      end
    end
  end
endgenerate

endmodule

module mux_clock_divider
#(parameter DIV_VALUE = 50000)
(
  input clk_in,
  input reset_n,
  output reg clk_out
);

generate
  if (DIV_VALUE <= 1) begin : passthrough
    always @(posedge clk_in, negedge reset_n)
    begin
      if (~reset_n)
        clk_out <= 0;
      else
        clk_out <= ~clk_out;
    end
  end
  else begin : divider
    localparam BITS = $clog2(DIV_VALUE) + 1;
    reg [BITS-1:0] counter;

    always @(posedge clk_in, negedge reset_n)
    begin
      if (~reset_n) begin
        counter <= 0;
        clk_out <= 0;
      end
      else if (counter >= (DIV_VALUE/2 - 1)) begin
        counter <= 0;
        clk_out <= ~clk_out;
      end
      else begin
        counter <= counter + 1;
      end
    end
  end
endgenerate

endmodule

module seven_seg_decoder(
```

```
  input [3:0] digit,
  output reg [6:0] seg
);

always @(*)
begin
  case(digit)
    4'd0: seg = 7'b1000000;
    4'd1: seg = 7'b1111001;
    4'd2: seg = 7'b0100100;
    4'd3: seg = 7'b0110000;
    4'd4: seg = 7'b0011001;
    4'd5: seg = 7'b0010010;
    4'd6: seg = 7'b0000010;
    4'd7: seg = 7'b1111000;
    4'd8: seg = 7'b0000000;
    4'd9: seg = 7'b0010000;
    default: seg = 7'b1111111;
  endcase
end

endmodule
```

Circuit Schematic



Pin Assignment

Conclusion

This project successfully implements a traffic light controller that meets all requirements: it stays green when no cars are present, uses 30-second green and 5-second yellow phases (with both lights yellow during transitions), and responds correctly to car sensors. Verified through simulation and tested on a Cyclone III FPGA with switches, LEDs, and a 7-segment display, the design works reliably in real time.